

Rosen — §8.2: Solving Linear Recurrence Relations

Selected Exercises

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Exercise 1

Determine which of these are linear homogeneous recurrence relations with constant coefficients. Also, find the degree of those that are.

(a) $a_n = 3a_{n-1} + 4a_{n-2} + 5a_{n-3}$

(b) $a_n = 2n a_{n-1} + a_{n-2}$

(c) $a_n = a_{n-1} + a_{n-4}$

(d) $a_n = a_{n-1} + 2$

(e) $a_n = a_{n-1}^2 + a_{n-2}$

(f) $a_n = a_{n-2}$

(g) $a_n = a_{n-1} + n$

Exercise 10

Prove Theorem 2.

Exercise 17

Prove this identity relating the Fibonacci numbers and the binomial coefficients:

$$f_{n+1} = \binom{n}{0} + \binom{n-1}{1} + \cdots + \binom{n-k}{k},$$

where n is a positive integer and $k = \lfloor n/2 \rfloor$. [*Hint:* Let $a_n = \binom{n}{0} + \binom{n-1}{1} + \cdots + \binom{n-k}{k}$. Show that the sequence $\{a_n\}$ satisfies the same recurrence relation and initial conditions satisfied by the sequence of Fibonacci numbers.]

Exercise 24

Consider the nonhomogeneous linear recurrence relation $a_n = 2a_{n-1} + 2^n$.

(a) Show that $a_n = n2^n$ is a solution of this recurrence relation.

(b) Use Theorem 5 to find all solutions of this recurrence relation.

(c) Find the solution with $a_0 = 2$.

Exercise 36

Let a_n be the sum of the first n perfect squares, that is, $a_n = \sum_{k=1}^n k^2$. Show that the sequence $\{a_n\}$ satisfies the linear nonhomogeneous recurrence relation $a_n = a_{n-1} + n^2$ and the initial condition $a_1 = 1$. Use Theorem 6 to determine a formula for a_n by solving this recurrence relation.

Exercise 38

- (a) Find the characteristic roots of the linear homogeneous recurrence relation $a_n = 2a_{n-1} - 2a_{n-2}$.
[Note: These are complex numbers.]
- (b) Find the solution of the recurrence relation in part (a) with $a_0 = 1$ and $a_1 = 2$.

Exercise 40

Solve the simultaneous recurrence relations

$$a_n = 3a_{n-1} + 2b_{n-1}, \quad b_n = a_{n-1} + 2b_{n-1}$$

with $a_0 = 1$ and $b_0 = 2$.

Exercise 41

- (a) Use the formula found in Example 4 for f_n , the n th Fibonacci number, to show that f_n is the integer closest to

$$\frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2} \right)^n.$$

- (b) Determine for which n , f_n is greater than $\frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2} \right)^n$ and for which n , f_n is less than $\frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2} \right)^n$.

Exercise 44

(Linear algebra required) Let A_n be the $n \times n$ matrix with 2s on its main diagonal, 1s in all positions next to a diagonal element, and 0s everywhere else. Find a recurrence relation for d_n , the determinant of A_n . Solve this recurrence relation to find a formula for d_n .

Exercise 45

Suppose that each pair of a genetically engineered species of rabbits left on an island produces two new pairs of rabbits at the age of 1 month and six new pairs of rabbits at the age of 2 months and every month afterward. None of the rabbits ever die or leave the island.

- (a) Find a recurrence relation for the number of pairs of rabbits on the island n months after one newborn pair is left on the island.
- (b) By solving the recurrence relation in (a) determine the number of pairs of rabbits on the island n months after one pair is left on the island.

Exercise 48

Show that the recurrence relation

$$f(n)a_n = g(n)a_{n-1} + h(n),$$

for $n \geq 1$, and with $a_0 = C$, can be reduced to a recurrence relation of the form

$$b_n = b_{n-1} + Q(n)h(n),$$

where $b_n = g(n+1)Q(n+1)a_n$, with

$$Q(n) = \frac{f(1)f(2) \cdots f(n-1)}{g(1)g(2) \cdots g(n)}.$$

Exercise 50

It can be shown that C_n , the average number of comparisons made by the quick sort algorithm when sorting n elements in random order, satisfies the recurrence relation

$$C_n = n + 1 + \frac{2}{n} \sum_{k=0}^{n-1} C_k$$

for $n = 1, 2, \dots$, with initial condition $C_0 = 0$.

- (a) Show that $\{C_n\}$ also satisfies the recurrence relation $nC_n = (n+1)C_{n-1} + 2n$ for $n = 1, 2, \dots$
- (b) Use Exercise 48 to solve the recurrence relation in part (a) to find an explicit formula for C_n .

Exercise 51

Prove Theorem 4.

Exercise 52

Prove Theorem 6.

Exercise 53

Solve the recurrence relation $T(n) = nT(n/2)^2$ with initial condition $T(1) = 6$ when $n = 2^k$ for some integer k . [*Hint*: Let $n = 2^k$ and then make the substitution $a_k = \log T(2^k)$ to obtain a linear nonhomogeneous recurrence relation.]